

1 Model

For our example, we adapt the model from Wolkovich & Donahue 2020, which is itself an adaptation of a common model of community dynamics, often conceptualized for annual plants with a seedbank. A more detailed description of the model can be found in Wolkovich & Donahue 2020.

We model a pair of competing species in which the seedbank (N) of species i at time $T + 1$ (where T is the beginning of an annual growing season) is determined by the survival (s) of ungerminated seeds from the previous year ($1 - g_i(T)$) plus new biomass (B_i) produced during the growing season (δ), which is converted to seeds at rate ϕ :

$$N_i(T + 1) = sN_i(T)(1 - g_i(T)) + \phi B_i(T + \delta) \quad (1)$$

Within season dynamics of biomass are modeled as R^* competition that begins when resources become available at the beginning of the growing season, $R(t = 0) = R_0$, where t indicates within-season time. For each species, biomass increases through germination ($G_i(t)$) and resource uptake ($f(R)$), which is converted into biomass at a species-specific rate c_i , less maintenance costs (m) (Fig:??b,c). Resource uptake per unit biomass is defined by the change in uptake per unit resource (a) and the maximum uptake ($1/u$).

$$\frac{dB_i}{dt} = (c_i f(R) - m) B_i + G_i(t) \quad (2)$$

$$f(R) = \frac{aR}{1 + auR} \quad (3)$$

The resource (R) itself declines across a growing season (Fig:??a) due to uptake by the standing biomass of both species and abiotic loss (ϵ):

$$\frac{dR}{dt} = - \sum_{i=1}^2 f(R) B_i - \epsilon R \quad (4)$$

Our model departs from similar community models by explicit modeling of within-season germination phenology: germination, ($G_i(t)$), is modeled as a time-dependent forcing function that adds daily biomass (germinants) according to a Hill equation where the timing of 50%germination ($\tau_{g50,i}$)*dependsonoverwinterconditions(i.e., daysofchilling)andspecies-specificsensitivitoychilling.*

$$G_i(t) = N_0 s_i b_i g_i(t) \quad (5)$$

$$g_i(t) = g_{max} \left(\frac{t^h}{\tau_{g50,i}^h + \text{within-season-time}^{nH}} \right) \quad (6)$$

where N_0 is the number of overwintering seeds, s_i is overwinter seed survival, b_i converts germinating seeds into biomass, and $g_i(t)$ is the percent germination at time (t) (Fig:??e,f). The Hill Equation defining $g_i(t)$ has a fixed Hill coefficient (h) and fixed maximum germination (g_{max}), and $\tau_{g50,i}$ is a Poisson random variable for the species-specific delay due to chilling (Figure S2a,b)

$$\tau_{g50,i} \sim \text{Poisson}(\tau_{delay,i}) \quad (7)$$

$$\tau_{delay,i} = \tau_{max} e^{-\tau_{\xi,i} \xi} \quad (8)$$

where ξ is the days of overwinter chilling in year T , $\tau_{\xi,i}$ is the species-specific sensitivity to chilling, and τ_{max} is the mean delay when overwinter chilling is zero. That is, germination depends on the interaction of fast and slow germination. ADD: While explaining the model, will need to reference (and include in figures), the plots that are made in `Prieff_Species.R` to visualize.

Simulations:

Given an overview of simulations performed, be sure to mention: two-species communities, species vary in *what* and explain a little what that does (for example: We varied species' resource use efficiency (*via* c_i), yielding species with different R^* (a metric of resource competition where a lower R^* means a species can draw the resource down to a lower level and is thus considered the superior competitor)) and then state something like "All other parameters were identical between species (Table S1)." Also need: starting conditions (for example: both species were initialized with a census size of $N(0) = 100$ per unit area, and the temporally varying parameters $R_0(t)$ and $\tau_p(t)$ were generated) and then some re-wording of this:

Within-year R^* competition dynamics were solved using an ode solver (`ode` in the R package `deSolve`) and ended when the resource was drawn down to $\min(R^*)$, i.e. the R^* value of the better resource competitor. The end-of-season biomass of each species was converted to seeds, and the populations were censused. At each census, a minimum cutoff was applied to define extinction from the model. Note that 'coexistence' in this model is defined by joint persistence through time and not by low density growth rate.

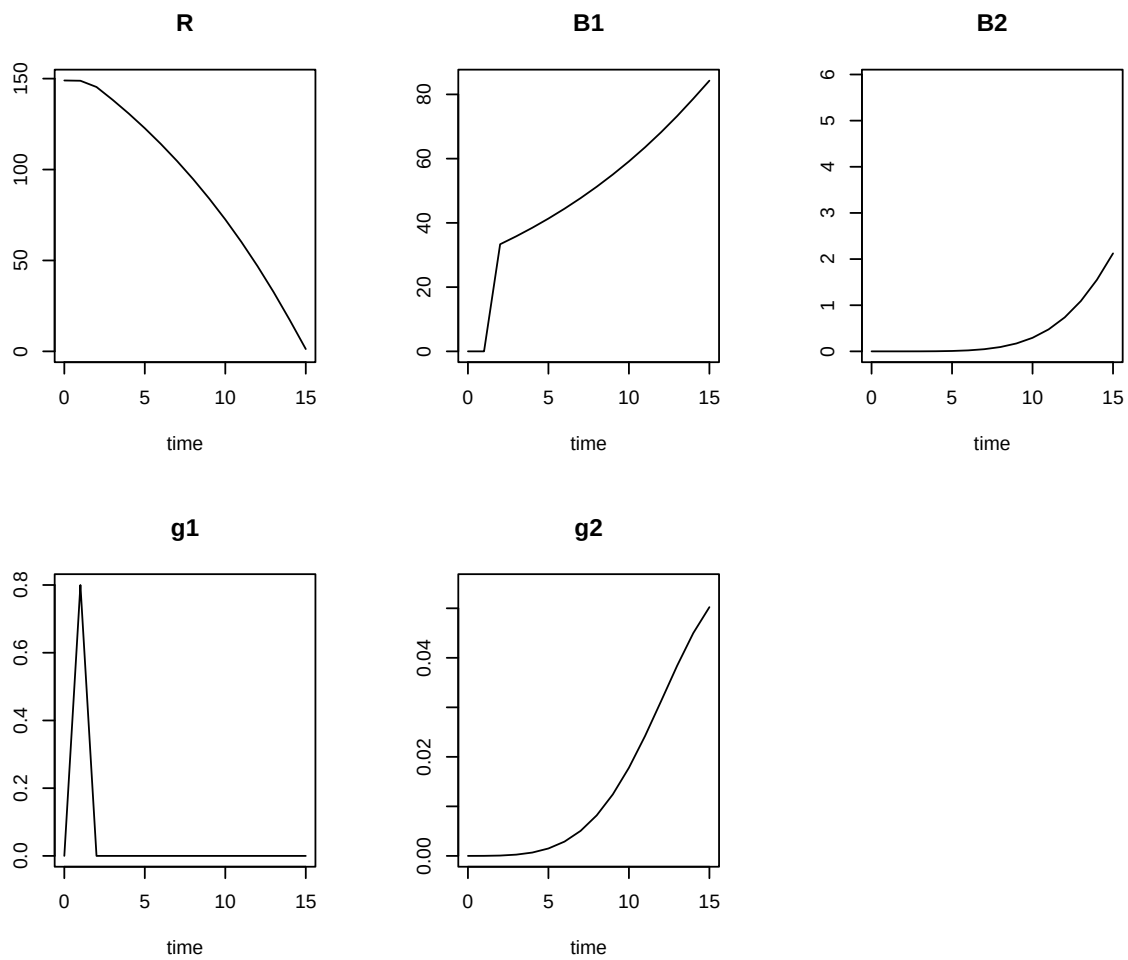


Figure S1: Caption here

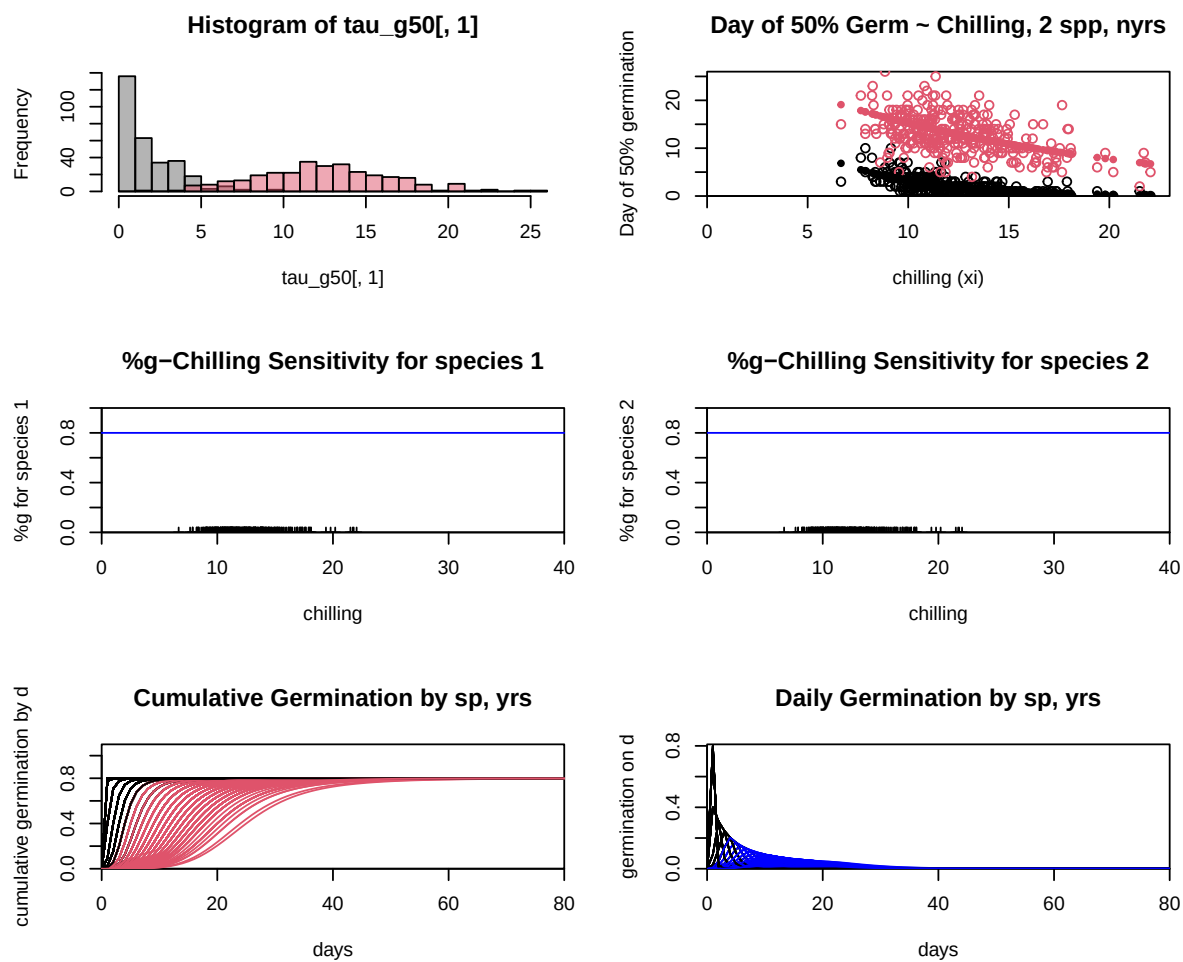


Figure S2: Caption here

Table S1: Parameter values, definitions, and units.

Parameter	Value(s)	Definition	Unit
N_i	initial conditions $N_i(0) = 100; \min(N_i(t)) = 10^{-4}$	census of seedbank of species i	seeds per unit area
s	0.8	survival of species i	unitless
B_i	see Eqn 2	biomass of species i	biomass
R_0	$\sim \log N(\mu, \sigma) \mu = \log(2), \sigma = 0.2$	annually varying initial value of resource at the beginning of the growing season	resource
ξ_0	$\sim ??$	chilling	chilling
c_i	$\sim \text{Unif}(8, 20)$	conversion efficiency of R to biomass of species i	$\frac{\text{biomass}}{\text{resource}}$
m	0.05	maintenance costs during growth season i	days^{-1}
a	20	uptake increase as R increases for species i	days^{-1}
u	1	inverse of maximum uptake for species i	$\frac{(\text{days})(\text{biomass})}{\text{resource}}$
θ	1	shape of uptake for species i	unitless
ϕ	0.05	conversion of end-of-season biomass to seeds	biomass^{-1} , but conceptually $\frac{\text{seeds}}{(\text{biomass})(\text{seeds})}$
ϵ	1	abiotic loss of R	days^{-1}
g_{max}	0.5	maximum germination rate of species	unitless
h	100	controls the the rate at which germination declines as τ_p deviates from optimum for species i	days^{-2}
g_i	see Eqn 6	germination rate	unitless
τ_g	$\sim \beta(?, ?)$	days of delay (given on weeks of chilling)	days
τ_{g50}	$\sim \text{Pois}(\tau_g)$	??	days
ξ_τ	$\sim ?, ?$	rate of delay (given weeks of chilling) i	days/week
b_i	1	biomass of a seedling	$\frac{\text{biomass}}{\text{seeds}}$
$f(R)$	see Eqn 3	resource uptake rate for species i	$\frac{\text{resource}}{(\text{days})(\text{biomass})}$
t	1	annual timestep	years
$0 \rightarrow \delta$	determined by rate of resource depletion	time during the growing season	days